

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

⚠ WARNING ⚠

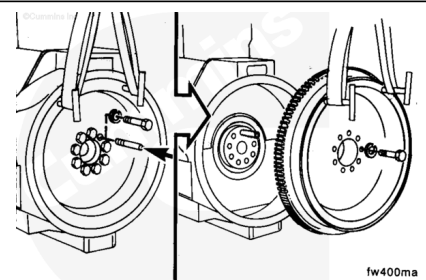
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift the component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air starter to prevent accidental engine starting. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.

Remove

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Remove two capscrews and install two guide studs.

Install the tee-handles.

Install the lifting straps.

Remove the remaining capscrews.

A mallet can be used to tap the flywheel, if necessary.

Remove the flywheel.

Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

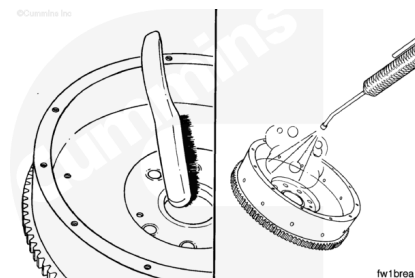
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the flywheel.

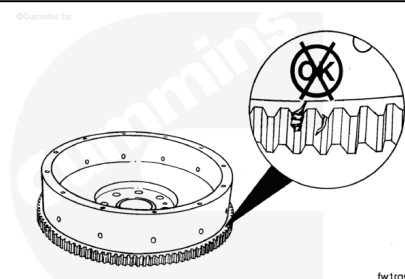
Use a wire brush to clean the crankshaft pilot bore.

Dry with compressed air.



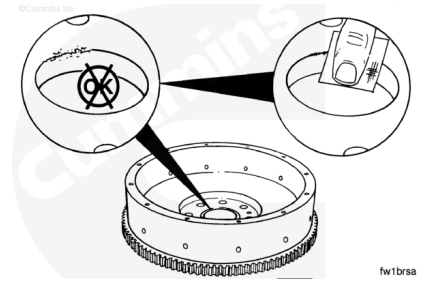
Inspect the teeth of the ring gear for wear or damage.

If the teeth of the ring gear are worn or damaged, the ring gear **must** be replaced.



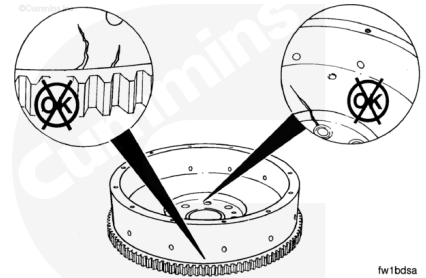
Use the dye penetrant, Part Number 3375452, to check the area of the mounting capscrew holes for cracks. There **must not** be any cracks in the flywheel mounting area.

Inspect the mounting pilot hole for damage. If the pilot hole is damaged, the flywheel **must** be replaced.



⚠ WARNING ⚠

Cummins Inc. does not recommend resurfacing the flywheel. Resurfacing the flywheel can cause it to be too thin and cause personal injury, property damage, or result in breakage.



Inspect the face of the flywheel for cracks or other damage.

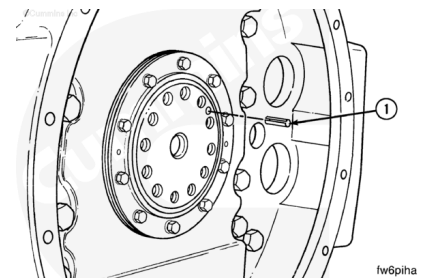
The flywheel **must** be replaced if cracked or otherwise damaged.

Install

Install the roll pin (1) in the crankshaft.

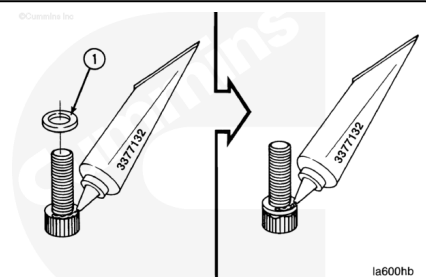
The pin is required to align the timing marks that are on the flywheel.

Note : The flywheel and flexplate are not the same for K38 and K50 engines. The valve and injector adjustment marks that are stamped on the parts are different.



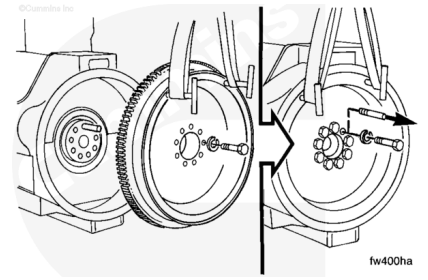
Lubricate the flywheel mounting capscrew threads and flange head surfaces with clean engine oil.

Install the washer onto the capscrew and lubricate the washer (1).



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Install two guide studs to prevent the flywheel from rotating.

Install the flywheel.

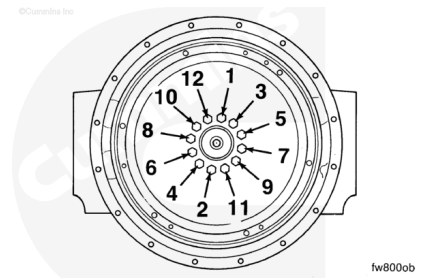
Install the flywheel mounting capscrews.

Remove the two guide studs and install the last two flywheel mounting capscrews.

Tighten the flywheel mounting capscrews in the sequence shown.

Torque Value:

1. 150 n•m [110 ft-lb]
2. 285 n•m [210 ft-lb]
3. 515 n•m [380 ft-lb]



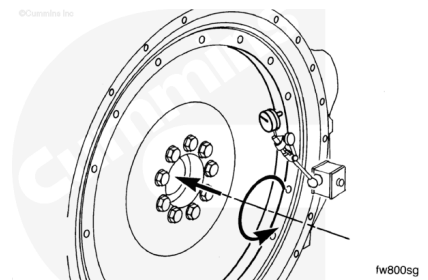
Measure the flywheel total indicator runout.

Note : The crankshaft end clearance must be pushed or pulled in the same direction each time a point is measured.

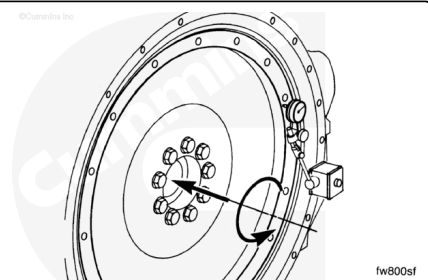
Attach an indicator as shown. Continue to look at the indicator while rotating the engine.

Maximum radial runout is 0.13 mm [0.005 inch]. If the runout is **not** within specification, the pilot on the flywheel is **not** positioned correctly on the crankshaft.

If the pilot is damaged, the flywheel **must** be replaced.



The maximum allowable flywheel face runout is determined by the distance from the center of the flywheel to the contact point of measurement, divided by 1000. Measure the flywheel face total runout.



Flywheel Face Total Indicator Runout

Number 0 (with a 254 mm [10 in] center distance)

	mm	in
MAX	0.254	0.010

If the flywheel alignment is **not** within specifications, check for interference between the flywheel and the crankshaft.

Finishing Steps

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WARNING

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- Install the drive unit and related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Connect the air starter. Refer to Procedure 012-022 in Section 12.
(/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Start the engine and check for proper operation.